

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		A – Oxygen – Diffusion/ simple diffusion B – Glucose - Facilitated diffusion/ active transport 1 mark for correct substances 1 mark for transport mechanisms	1	1		2		2
		(ii)		• {Oxygen/ (substance) A} passes through {phospholipid/ bilayer} because it is non-polar (1) • {Line A/ oxygen/ diffusion}: {linear relationship/ positive correlation/ directly proportional} (between concentration gradient and rate of diffusion)/ or description (1) • {Glucose/ (substance) B} uses {channel/ carrier/ intrinsic/ transmembrane} proteins as it is polar (1) • {Line B/ glucose/ facilitated diffusion/ active transport}: the rate is eventually limited by the number of {channel/ carrier/ intrinsic/ cell membrane} proteins (1) Must refer to carrier proteins if use active transport	2	2		4		
	(b)	(i)		Mark points in pairs • No {DNA/ genes} (1) • to code for haemoglobin (1) OR • No ribosomes (1) • for translation/ protein synthesis. (1) OR • No Golgi {body/ apparatus} (1) • so no post-translational modification (1) OR • No mitochondria (1) • to make ATP for protein synthesis (1)			2	2		

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		(ii)		No mitochondria (1) No ATP for Active transport (1)			2	2		
				Question 3 total	3	3	4	10	0	2